

BAYANBULAK, China

WMO: 515420

Lat: 43.0336N

Lon: 84.1491E

Elev: 2459

StdP: 75.07

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.1	-35.8	-41.2	0.1	-38.0	-38.8	0.1	-35.7	11.4	-16.4	9.2	-16.4	0.5	340	0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	21.4	10.9	20.0	10.3	18.6	9.8	12.3	18.9	11.4	17.8	10.7	16.8	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	10.1	14.3	8.7	9.5	13.5	7.9	9.0	13.0	42.7	19.0	40.1	18.1	37.9	16.9	16.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	9.9	8.4	DB	-40.1	24.7	3.3	1.5	-42.5	25.7	-44.4	26.6	-46.2	27.4	-48.6	28.5	
			WB	-40.1	13.8	3.0	1.1	-42.3	14.6	-44.0	15.3	-45.7	15.9	-47.9	16.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.3	-25.6	-21.0	-9.5	1.5	6.2	9.9	11.7	10.8	6.6	0.0	-9.9	-21.1
	DBStd	13.85	5.86	6.13	6.77	4.45	3.25	2.50	2.24	2.51	3.12	4.50	6.65	5.91
	HDD10.0	4990	1104	869	604	256	121	31	8	19	107	311	597	963
	HDD18.3	7890	1362	1102	863	506	375	252	204	234	353	569	847	1221
	CDD10.0	143	0	0	0	0	4	28	62	44	4	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	12	0	0	0	0	0	0	10	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	2.9	1.5	1.9	2.9	4.1	3.9	3.4	3.2	3.0	3.1	3.1	2.5	1.8
Precipitation	PrecAvg	265	2	3	4	10	30	60	62	51	28	5	3	3
	PrecMax	355	12	10	17	21	66	148	160	110	60	15	19	9
	PrecMin	212	0	0	0	3	10	24	20	13	9	0	0	0
	PrecStd	36	3	2	4	5	14	27	27	23	14	4	4	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.4	-3.8	8.4	17.7	19.9	21.3	24.7	23.0	20.6	15.0	7.6	-5.1
		MCWB	-9.6	-6.3	1.4	6.3	8.0	10.2	12.1	11.6	9.0	5.3	0.8	-6.8
	2%	DB	-11.6	-7.1	5.1	14.6	17.6	19.6	21.8	21.2	18.3	12.1	4.2	-7.9
		MCWB	-12.5	-8.7	0.3	5.0	7.3	9.7	11.7	10.9	8.4	4.2	-0.9	-9.3
	5%	DB	-14.2	-9.5	2.4	11.9	15.8	18.2	20.0	19.6	16.5	10.0	1.5	-10.0
		MCWB	-15.1	-10.6	-1.3	3.8	6.7	9.3	11.1	10.4	7.7	3.1	-2.0	-11.1
	10%	DB	-16.9	-11.8	0.2	9.6	13.8	16.5	18.2	17.9	14.6	7.9	-1.2	-12.4
		MCWB	-17.5	-12.7	-2.4	2.9	5.9	8.8	10.6	10.0	6.9	2.0	-3.8	-13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.5	-6.1	2.6	7.3	9.4	11.8	14.0	13.1	10.1	6.1	1.8	-7.0
		MCDB	-8.5	-4.1	7.2	15.5	17.0	18.1	20.2	19.9	18.3	13.1	5.5	-5.4
	2%	WB	-12.5	-8.7	0.6	5.6	8.3	10.8	12.7	12.0	9.0	4.8	-0.4	-9.2
		MCDB	-11.6	-7.2	4.1	12.9	15.5	17.0	19.6	18.7	16.6	11.0	3.6	-7.9
	5%	WB	-15.0	-10.6	-0.9	4.4	7.4	10.1	11.8	11.2	8.1	3.7	-2.1	-11.1
		MCDB	-14.2	-9.5	2.1	10.6	14.1	16.1	18.2	17.6	15.1	9.0	1.4	-9.9
	10%	WB	-17.5	-12.7	-2.2	3.3	6.5	9.4	11.0	10.4	7.3	2.5	-3.8	-13.2
		MCDB	-16.9	-11.9	0.3	8.6	12.6	15.1	17.2	16.5	13.7	7.0	-1.2	-12.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.5	12.0	12.0	12.1	11.2	11.6	12.7	13.8	13.7	11.3	10.0
		MCDBR	11.8	12.0	12.8	17.5	16.6	15.2	15.7	17.0	18.2	17.3	15.1	11.6
		MCWBR	10.8	10.6	8.4	8.9	7.9	6.8	7.1	8.0	9.2	9.8	10.1	10.2
	5% WB	MCDBR	11.8	11.9	11.5	14.4	13.9	12.1	13.2	13.8	15.9	15.2	14.5	11.6
		MCWBR	10.8	10.5	7.8	7.8	6.8	5.8	6.3	6.8	8.1	8.6	9.8	10.1
Clear-Sky Solar Irradiance	taub	0.239	0.275	0.326	0.367	0.362	0.373	0.355	0.348	0.331	0.298	0.264	0.234	
	taud	2.393	2.299	2.201	2.119	2.214	2.243	2.341	2.355	2.358	2.379	2.403	2.428	
	Ebn at Noon	922	928	912	895	909	896	909	906	898	889	876	899	
	Edh at Noon	75	99	126	149	139	136	122	116	107	91	75	66	
All-Sky Solar Radiation	RadAvg	2.28	3.27	4.38	5.16	5.73	5.83	5.87	5.47	4.55	3.28	2.27	1.88	
	RadStd	0.11	0.15	0.28	0.38	0.36	0.37	0.34	0.31	0.24	0.26	0.15	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page